Tim Richardson

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SUMMARY

Senior Software Engineer and Engineering Manager with 13+ years of experience building scalable data platforms, specialising in genomics and healthcare. Currently leading AI transformation initiatives and managing engineering teams at Genomics England. Strong track record delivering full-stack solutions across Python, Go, and cloud infrastructure, with deep expertise in bioinformatics pipelines and ML integration. Builds and publishes open-source tooling for AI coding agents, evaluated with pre-registered experiments.

TECHNICAL SKILLS

Languages: Python, Go, TypeScript, JavaScript, Rust, SQL
AI/ML: LLMs, RAG, LangChain, OpenAI API, Anthropic API, Google ADK, NumPy, SciPy, Pandas
AI Agents: Claude Code plugins and skills, agent memory, agent evaluation and benchmarking
Testing: pytest, mutation testing, coverage analysis, TDD, CI quality gates
Backend: FastAPI, Django, Node.js, GraphQL, REST APIs, Microservices
Frontend: React, Next.js, Redux, Svelte, Tauri, D3.js
Data: PostgreSQL, Elasticsearch, Redis, Vector Databases, Graph Databases, FHIR, Ontologies
Cloud & Infrastructure: AWS, GCP, Docker, Kubernetes, Terraform, CI/CD
Bioinformatics: Nextflow, nf-core, VEP, Samtools, BWA, Singularity, LSF

EXPERIENCE

Genomics England London, UK
Senior Software Engineer and Performance Development Lead Dec 2021 – Present

- **Engineering Leadership:** Line Manager for 6 direct reports, responsible for mentorship, performance management, and delivery oversight.
- **Workflow Modernisation:** Key role in migrating complex workflow orchestration systems to Nextflow, improving scalability and reproducibility.
- **AI Transformation:** Leading internal initiative to introduce and operationalize AI coding assistants across engineering, driving efficiency.
- **Backend Development:** Architecting and maintaining robust backend services using Python and AWS for high-throughput genomic data processing.

IQVIA London, UK
Senior Software Engineer - AI/ML Team Mar 2019 – Dec 2021
Stack: go, fastapi, typescript, reactjs, redux, elasticsearch, postgresql, d3.js

- **Full-Stack Development:** Built data analytics product for real-world healthcare data, developing backend services in Go and frontend in React.
- **AI Model Deployment:** Deployed machine learning models using FastAPI, enabling scalable inference for healthcare analytics.
- **Statistical Features:** Developed and implemented statistical features for the analytics product, enhancing data insights capabilities.
- Coordinating releases, product development, process improvement, code review.

Project Shivom London, UK
Lead Developer Oct 2018 – Mar 2019
Stack: blockchain, smartcontracts, google-genomics, google-cloud-build, postgresql, graphql, react-apollo

- Design, development, and deployment of a genomic data sharing solution with blockchain components.
- Managed development team, created wireframes, developed backend services (REST APIs) and React SPA frontend.
- Deployed using Docker and App Engine.

Polecat London, UK
Python Contractor Jul 2018 – Oct 2018
Stack: django, django-rest-framework, graphene-python, reactjs, elasticsearch, postgresql, plotly

- Built a Python reporting service to generate PowerPoint marketing reports.
- Engaged with technical teams and product managers to produce software specifications and wire-frames.
- Solution built using Python, Django, GraphQL, React, Elasticsearch, and Plotly.js.

Congenica Hinxton, UK
Senior Frontend Developer (React) Mar 2016 – Jul 2018
Stack: perl, reactjs, react-router, redux, reactive-programming, bioperl

- Developed new and optimised existing features for clinical genomics tool using Perl APIs and React frontend.
- Focused on optimal data flow for data-intensive UI.
- Led development of an API to integrate Congenica's Sapientia platform with Genomics England's 100k genomes project.

Repositive Ltd London, UK
Lead Developer Nov 2014 – Mar 2016
Stack: docker, node.js, ansible, ember.js, git, agile, postgresql, elasticsearch

- Led development of a genomic data discovery platform, including infrastructure and development processes.
- Developed web app architecture based on Node.js Express API using Redis, Postgres, and Elastic-search.
- Built UI as an Ember.js single-page application.

Wolfson Institute for Biomedical Research London, UK
Freelance Bioinformatics Project Oct 2014 – Nov 2014

- Analyzed ChIP-SEQ project: trimming and QC, peak calling, motif discovery, GO annotation.

Fire Tech Camp London, UK
Instructor Jun 2014 – Aug 2014

- Taught Python programming to children aged 11-15.

Ixxy Brighton/Remote
Django Developer Mar 2013 – May 2014

- Built and maintained sites based on Django/Python CMS system.

Ecomarket.com London, UK
Django Developer Jan 2013 – Mar 2013

- Backend developer for an online marketplace using Django, Python, and PostgreSQL.

Swerve Concepts London, UK
Junior Software Developer Feb 2011 – Jan 2013

- Full stack developer using Python and PHP frameworks. Developed cloud-based products Swerve Media Hub and SignUpAnywhere.

SELECTED OPEN SOURCE

Agentic Verification (github.com/timini/agentic-verification)
Claude Code plugin and GitHub Action that checks whether tests catch broken code, by running every mutant of the lines a change touches. Backed by a pre-registered study and agent benchmarks on 8 open-source Python libraries, published as a paper: on 14 real commits, the agent's tests caught 99% of mutants with the plugin against 92% without.

drill-me (github.com/timini/drill-me)

Claude Code plugin that turns Claude into an adaptive tutor: it quizzes before it explains, schedules reviews with spaced repetition and remembers what you got wrong across sessions.

neuromem (github.com/timini/neuromem)
Long-term memory for LLM agents modelled on hippocampal consolidation, with providers for Gemini, OpenAI, Claude and local models. Scores 0.870 on the LongMemEval-s benchmark (n=100).

horn-simulation (github.com/timini/horn-simulation)
Horn loudspeaker design pipeline: generates geometries, solves the Helmholtz equation with FEM, matches real drivers and ranks the combinations.

Crate Digger (github.com/timini/crate-digger)
Local-first desktop app for DJs that discovers, downloads, analyses and ranks new music. Rust, Tauri and Svelte, with a Rust metadata service on Cloud Run.

Wiggle Room (github.com/timini/wiggleroom)
45 VCV Rack 2 synthesiser modules built with Faust physical-modelling DSP.

rewire.it (rewire.it)
Personal blog on software engineering, genomics, and technology.

EDUCATION

Birkbeck, University of London — London, UK
MSc Bioinformatics & Systems Biology (Merit) — Sep 2012 – Sep 2014

- Studied statistics, protein structure, sequence alignment, systems biology, and software development.
- Thesis: Tracing human migration using MCMC methods for population clustering with dense genotype data.

Leeds University — Leeds, UK
BSc Cognitive Science (Hons) — Sep 2005 – Sep 2008

- Interdisciplinary course covering computing, psychology, and philosophy.
- Dissertation: Data mining techniques for dynamic data sets.